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# PROG 102: PRINCIPLES OF SOFTWARE ENGINEERING

**ASSIGNMENT TITTLE:** **FINAL PROJECT**
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**LECTURER:** **MR. SANTIGIE FODAY KAMARA**
**CLASS:** **BSEM1203F**
**SEMESTER/YEAR:** **SEMESTER 1/ YEAR 1**

**Academic Honesty Policy Statement**

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# INTRODUCTION

## 1.1 Background

Access to suitable rental accommodation remains a significant challenge in many urban communities. Prospective tenants often spend considerable time searching for properties that meet their requirements, while landlords and property agents struggle to identify suitable tenants efficiently. The rental process frequently relies on social media advertisements, personal referrals, and phone calls, resulting in fragmented information and slow communication.

Digital technologies provide opportunities to improve the rental housing process through centralized information management and automated matching systems. RentSure is proposed as a Digital Public Good that connects tenants, landlords, and property agents through a web-based platform. The system enables tenants to submit accommodation requirements and allows landlords or agents to publish available properties. A matching mechanism then recommends suitable properties based on user requirements.

## 1.2 Problem Statement

The current rental housing process faces several challenges:

- Difficulty locating suitable accommodation.
- Lack of centralized housing information.
- Time-consuming manual matching.
- Inefficient communication between tenants and landlords.
- Limited transparency in the rental process.

These issues increase the time and effort required to secure accommodation.

## 1.3 Aim

The aim of this project is to design a Digital Public Good that improves the rental housing process through automated tenant-property matching.

## 1.4 Objectives

The objectives of the project are:

- To create a platform for tenants and landlords.
- To automate tenant-property matching.
- To improve housing accessibility.
- To reduce rental search time.
- To demonstrate software engineering principles.
- To support Sustainable Development Goal 11.

# SOFTWARE DEVELOPMENT LIFE CYCLE

## 2.1 Selected SDLC Model

The Agile Software Development Life Cycle was selected for the development of RentSure. Agile supports iterative development, continuous improvement, and rapid feedback.

## 2.2 Requirements Phase

Requirements were gathered through observation of existing rental practices and analysis of MVP 2 validation findings. Key requirements included user authentication, requirement management, property listing management, and matching functionality.

## 2.3 Design Phase

The design phase involved:

- User interface design using Figma.
- System architecture design.
- Database planning.
- Matching workflow design.

## 2.4 Development Phase

Development was divided into multiple iterations. MVP 2 validated the matching concept using Local Storage, while MVP 3 extends the system with authentication and database support.

## 2.5 Testing Phase

Testing focused on:

- Navigation functionality.
- CRUD operations.
- Matching logic.
- Data persistence.

## 2.6 Deployment Phase

The final solution is intended to be deployed as a web application accessible through mobile and desktop browsers.

## 2.7 Maintenance Phase

Future improvements include messaging, push notifications, and AI-assisted recommendations.

# SYSTEM ARCHITECTURE DESIGN

## 3.1 Architecture Overview

RentSure follows a three-tier architecture.

### Presentation Layer

The presentation layer consists of mobile-friendly web interfaces used by tenants and landlords.

### Business Logic Layer

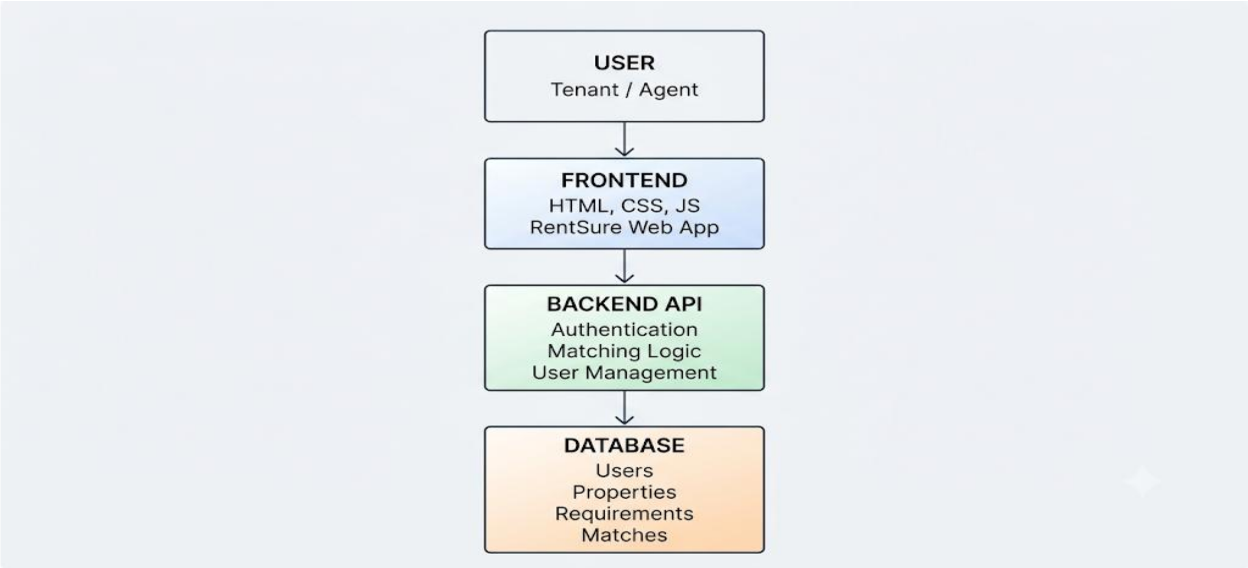
The business logic layer handles:

- User authentication
- Property management
- Requirement management
- Matching calculations

### Data Layer

The data layer stores users, properties, requirements, and match results in a MySQL database.

## 3.2 Architecture Diagram



### 3.3 Component Interaction

A tenant creates a housing requirement. The matching engine compares the requirement against available properties and generates compatibility scores. Matching results are then displayed to the user through the web interface.

# SYSTEM ANALYSIS

## 4.1 Existing System

Current rental processes commonly rely on:

- Social media advertisements
- Phone calls
- Personal referrals
- Physical property visits

## 4.2 Problems Identified

Major limitations include:

- Slow communication
- Inaccurate information
- Manual matching
- Lack of centralized records

## 4.3 MVP 2 Validation

A preliminary MVP 2 prototype was developed to validate the matching concept.

Features included:

- Tenant management
- Property management
- Matching engine
- Local Storage persistence

## 4.4 Findings

### What Worked

- Matching recommendations were useful.
- Navigation was simple.
- Data entry workflows were straightforward.

### Limitations

- Single-user operation
- No authentication
- No centralized database
- Not suitable for deployment

These findings informed the design of MVP 3.

# PROTOTYPE DEVELOPMENT

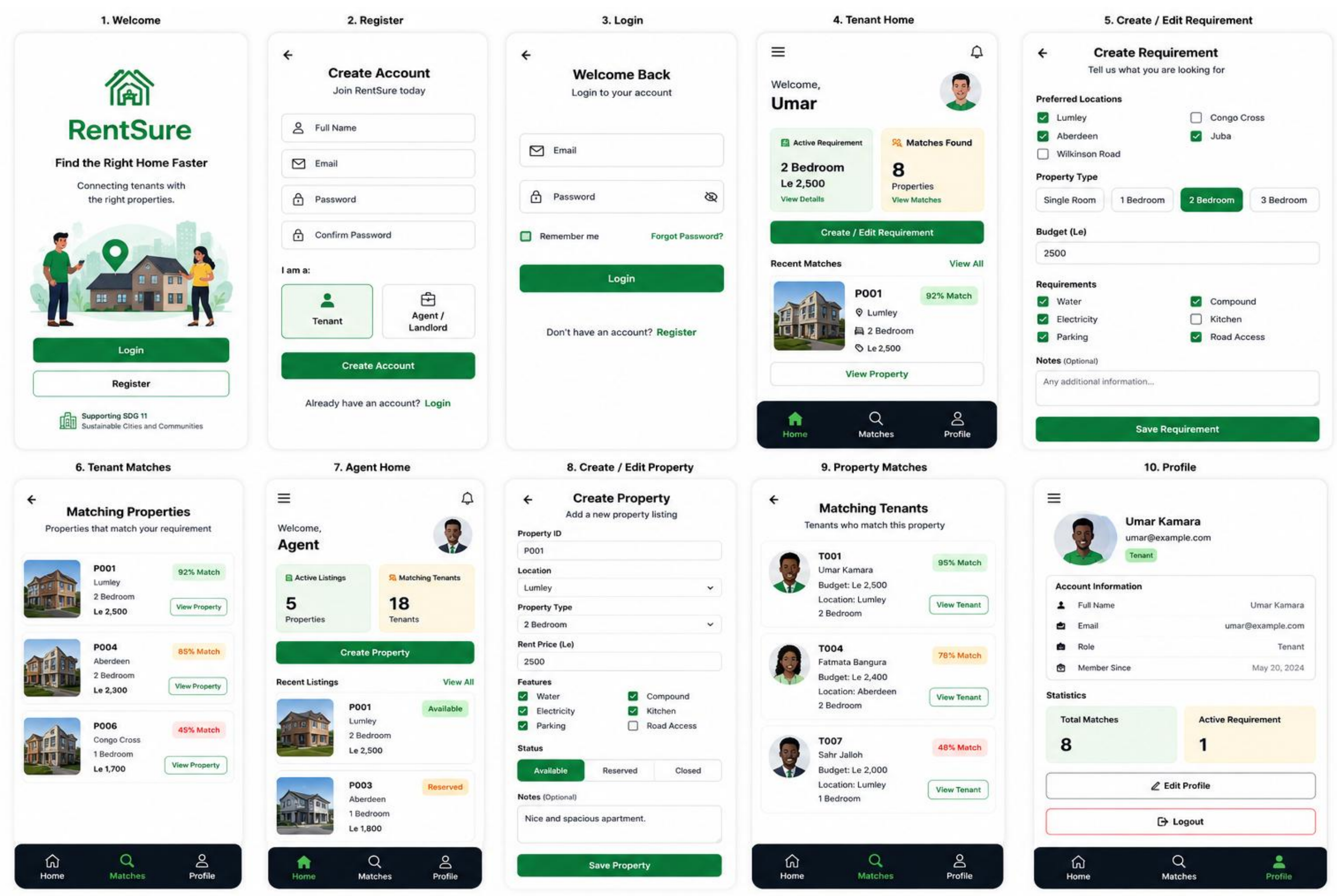
## 5.1 Prototype Tool

Figma was used to design the user interface.

## 5.2 Key Screens

The prototype includes:

1. Welcome Screen
2. Login Screen
3. Registration Screen
4. Tenant Dashboard
5. Matching Properties
6. Property Details
7. Create Requirement
8. Agent Dashboard
9. Create Property
10. Profile Screen



## 5.3 Design Principles

The interface follows mobile-first design principles:

- Simplicity
- Consistency
- Accessibility
- Easy navigation

# GIT AND GITHUB INTEGRATION

## 6.1 Purpose of GitHub

GitHub was used for:

- Version control
- Project backup
- Documentation management
- Collaboration

## 6.2 Repository Contents

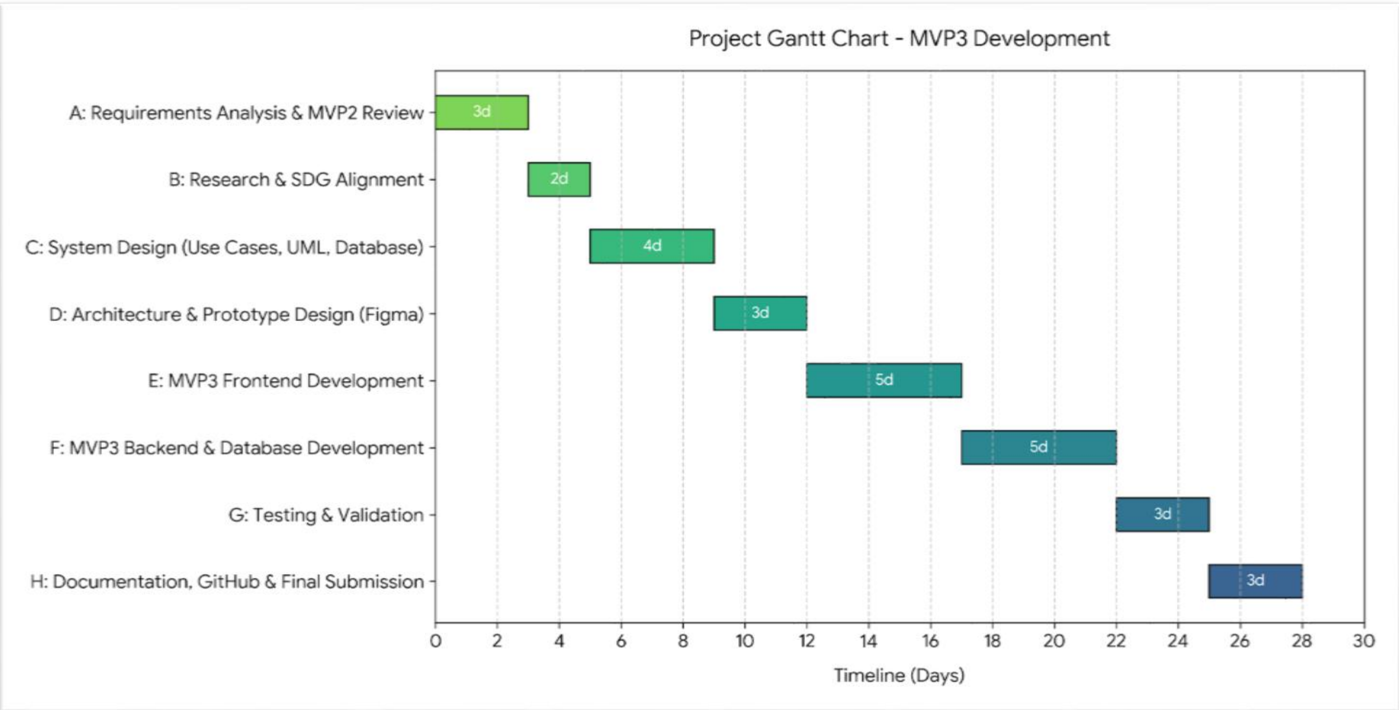
The repository contains:

- Source code
- Project documentation
- Diagrams
- Prototype assets

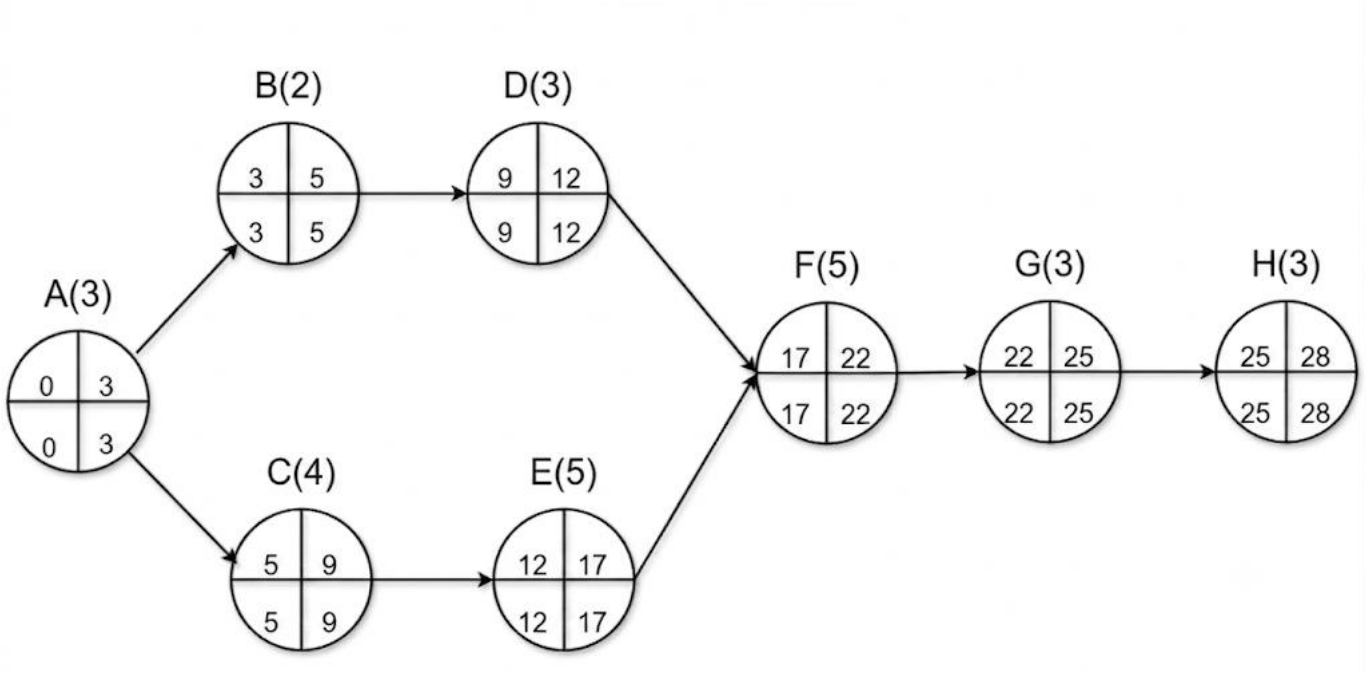
# PROJECT MANAGEMENT

## 7.1 Gantt Chart

| ID | Activity | Duration (Days) | Dependency |
|---|---|---|---|
| **A** | Requirements Analysis & MVP2 Review | 3 Days | None |
| **B** | Research & SDG Alignment | 2 Days | Requirements |
| **C** | System Design (Use Cases, UML, Database) | 4 Days | Research |
| **D** | Architecture & Prototype Design (Figma) | 3 Days | System Design |
| **E** | MVP3 Frontend Development | 5 Days | Prototype Design |
| **F** | MVP3 Backend & Database Development | 5 Days | Frontend Development |
| **G** | Testing & Validation | 3 Days | Backend Development |
| **H** | Documentation, GitHub & Final Submission | 3 Days | Testing |



## 7.2 Network Diagram



# LICENSE, DESIGN QUALITY AND SDG RELEVANCE

## 8.1 License Selection

The MIT License was selected because it promotes open-source collaboration while remaining easy to understand and implement.

## 8.2 Design Quality

The user interface was designed using mobile-first principles and emphasizes simplicity, consistency, and usability.

## 8.3 SDG Alignment

RentSure supports Sustainable Development Goal 11: Sustainable Cities and Communities.

The system contributes by:

- Improving access to housing information.
- Reducing rental search time.
- Increasing transparency in housing markets.
- Supporting digital transformation.

## CONCLUSION

RentSure is a Digital Public Good designed to improve the rental housing process through automated tenant-property matching. The project demonstrates the application of software engineering principles including Agile development, system architecture design, prototyping, project management, and version control.

Validation through MVP 2 confirmed the usefulness of the matching concept and informed the design of MVP 3. The proposed system provides a foundation for improving housing accessibility while contributing to Sustainable Development Goal 11.

Future enhancements may include real-time messaging, push notifications, and AI-assisted recommendations.

## REFERENCES


1. PROG102 Principles of Software Engineering Lecture Notes, 2026.
2. Agile Alliance Documentation.
3. Figma Documentation.
4. GitHub Documentation.
5. United Nations Sustainable Development Goals.